

Ex. No: 6 — Use Sleuth Kit to Analyze Digital Evidence

Aim

To analyze a forensic disk image using The Sleuth Kit and examine the partition structure, file system, files, metadata, and recover a file from the digital evidence.

Tools Required

- The Sleuth Kit 4.14.0
- Windows Command Prompt
- Windows File Explorer
- Forensic Disk Image:
 - 4Dell Latitude CPI.E01
 - 4Dell Latitude CPI.E02

Evidence Used

The forensic image files were stored in:

D:\DF\

The evidence consisted of:

4Dell Latitude CPI.E01

4Dell Latitude CPI.E02

Procedure

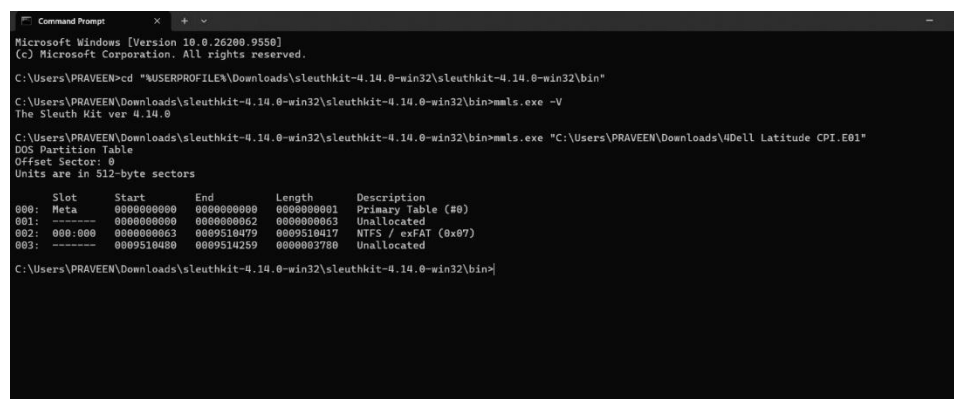
Step 1: Identify Partition Structure

The mmls command was used to identify the partition structure of the forensic disk image.

Command:

mmls "D:\DF\4Dell Latitude CPI.E01"

The output identified an NTFS/exFAT partition beginning at sector 63. Therefore, an offset of 63 was used for subsequent file-system analysis.



```
Microsoft Windows [Version 10.0.26200.9550]
(c) Microsoft Corporation. All rights reserved.

C:\Users\PRAVEEN>cd "%USERPROFILE%\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin"

C:\Users\PRAVEEN\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mmls.exe -V
The Sleuth Kit ver 4.14.0

C:\Users\PRAVEEN\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mmls.exe "C:\Users\PRAVEEN\Downloads\4Dell Latitude CPI.E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

   Slot   Start      End      Length  Description
   ----   -
000: Meta   0000000000  0000000000  0000000001 Primary Table (#0)
001: ----- 0000000000  0000000002  0000000003 Unallocated
002: 000:000 0000000003  0009510479  0009510417 NTFS / exFAT (0x07)
003: ----- 0009510480  0009514259  0000003780 Unallocated

C:\Users\PRAVEEN\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>
```

Fig. 1: Partition structure identified using mmls

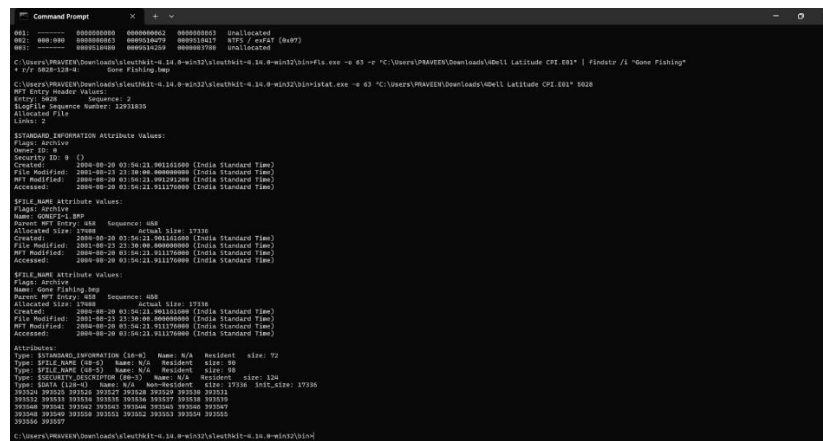
Step 2: Analyze File System

The fsstat command was used to obtain information about the file system.

Command:

```
fsstat -o 63 "D:\DF\4Dell Latitude CPI.E01"
```

The output identified the file system as NTFS. The analysis also provided information about the MFT, sector size, cluster size, and root directory.



```
Command Prompt
fsstat -o 63 "D:\DF\4Dell Latitude CPI.E01"

MFT Entry Header Values:
MFT Entry: 5028
Sequence: 2
Sample Sequence Number: 1291435
Allocated File:
Cluster: 2

$ATTRIBUTE_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 0
Created: 2008-08-20 01:50:21.001161000 (India Standard Time)
File Modified: 2008-08-20 21:38:00.000000000 (India Standard Time)
MFT Modified: 2008-08-20 01:50:21.001161000 (India Standard Time)
Accessed: 2008-08-20 01:50:21.001161000 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: GONE_FISHING.bmp
Parent MFT Entry: 5028
Sequence: 400
Allocated Size: 17936
Actual Size: 17936
Created: 2008-08-20 01:50:21.001161000 (India Standard Time)
File Modified: 2008-08-20 21:38:00.000000000 (India Standard Time)
MFT Modified: 2008-08-20 01:50:21.001161000 (India Standard Time)
Accessed: 2008-08-20 01:50:21.001161000 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: Gone Fishing.bmp
Parent MFT Entry: 400
Sequence: 400
Allocated Size: 17936
Actual Size: 17936
Created: 2008-08-20 01:50:21.001161000 (India Standard Time)
File Modified: 2008-08-20 21:38:00.000000000 (India Standard Time)
MFT Modified: 2008-08-20 01:50:21.001161000 (India Standard Time)
Accessed: 2008-08-20 01:50:21.001161000 (India Standard Time)

Attributes:
Type $ATTRIBUTE_INFORMATION (is-a) Name N/A Resident Size 72
Type $FILE_NAME (is-a) Name N/A Resident Size 80
Type $FILE_NAME (is-a) Name N/A Resident Size 80
Type $SECURITY_DESCRIPTOR (is-a) Name N/A Resident Size 128
Type $DATA (is-a) Name N/A Non-Resident Size 17936 Size_Bytes 17936
39352 39353 39354 39355 39356 39357 39358 39359
39360 39361 39362 39363 39364 39365 39366 39367
39368 39369 39370 39371 39372 39373 39374 39375
39376 39377

C:\Users\MAVERICK\Downloads\4Dell Latitude CPI.E01>
```

Fig. 2: NTFS file system information obtained using fsstat

Step 3: List Files and Directories

The fls command was used to recursively list files and directories in the forensic image.

Command:

```
fls -o 63 -r "D:\DF\4Dell Latitude CPI.E01"
```

The file Gone Fishing.bmp was identified in the output with MFT entry 5028.

```
r/r 5028-128-4: Gone Fishing.bmp
```

Fig. 3: Gone Fishing.bmp identified using fls

Step 4: Analyze File Metadata

The istat command was used to examine the metadata associated with Gone Fishing.bmp.

Command:

```
istat -o 63 "D:\DF\4Dell Latitude CPI.E01" 5028
```

The metadata showed:

- MFT Entry: 5028
- File Name: Gone Fishing.bmp

- **Actual Size: 17,336 bytes**
- **Allocated Size: 17,408 bytes**
- **Parent MFT Entry: 458**
- **File System: NTFS**

```
D:\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>icat -o 63 "D:\DF\4Dell Latitude CPI.E01" 5028 > "D:\DF\DF_Output\Gone_Fishing.bmp"
The system cannot find the path specified.

D:\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mkdir "D:\DF\DF_Output"

D:\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>icat -o 63 "D:\DF\4Dell Latitude CPI.E01" 5028 > "D:\DF\DF_Output\Gone_Fishing.bmp"

D:\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>|
```

Fig. 4: Metadata of Gone Fishing.bmp obtained using istat

Step 5: Recover the File

The icat command was used to recover Gone Fishing.bmp from the forensic disk image.

The recovered file was saved in the DF_Output folder.

Command:

```
icat -o 63 "D:\DF\4Dell Latitude CPI.E01" 5028 > "D:\DF\DF_Output\Gone_Fishing.bmp"
```

The recovered file was stored as:

D:\DF\DF_Output\Gone_Fishing.bmp

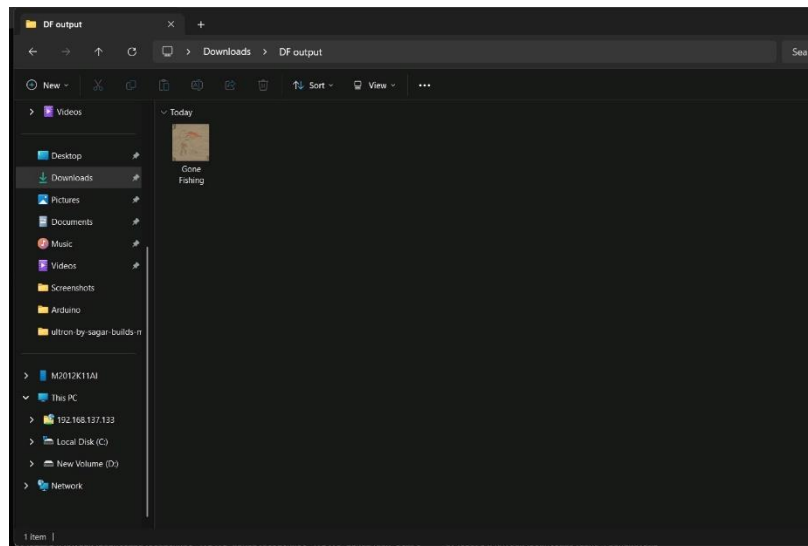


Fig. 5: Recovery of Gone Fishing.bmp using icat

Step 6: Verify the Recovered File

The recovered Gone_Fishing.bmp file was opened using Windows File Explorer. The image was displayed successfully, confirming that the file was recovered from the forensic disk image.

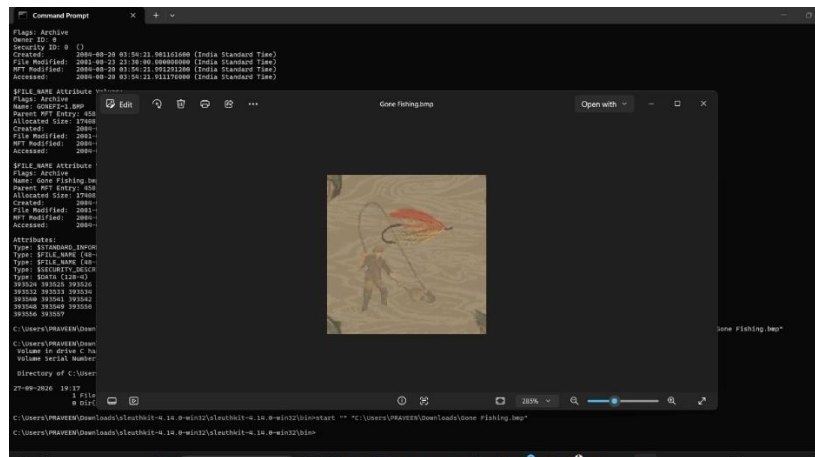


Fig. 6: Recovered Gone Fishing.bmp

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

The forensic disk image was successfully analyzed using The Sleuth Kit 4.14.0. The partition structure and NTFS file system were identified, files and directories were listed, and Gone Fishing.bmp was identified with MFT entry 5028. Its metadata was analyzed using istat, and the file was successfully recovered using icat.